

Question	Answer	Marks
7(a)	a <u>nucleus</u> containing <u>protons</u> and <u>neutrons</u>	B1
	(a nucleus) surrounded by (orbital) electrons	B1
7(b)(i)	alpha (decay)	B1
7(b)(ii)	beta-minus (decay)	B1
7(c)	ratio = $n \times \text{elementary charge} / \text{mass}$ (mass =) $(n \times) 1.60 \times 10^{-19} / 1.76 \times 10^{11} = 9.09 \times 10^{-31}$	B1
	OR (mass =) $(n \times) 1.60 \times 10^{-19} / (1.76 \times 10^{11} \times 1.66 \times 10^{-27}) = 0.000548$	(B1)
	OR (charge =) $1.76 \times 10^{11} \times 9.11 \times 10^{-31} = 1.60 \times 10^{-19}$	(B1)
	OR (ratio =) $1.60 \times 10^{-19} / 9.11 \times 10^{-31} = 1.76 \times 10^{11}$	(B1)
	(so X is a) positron	B1